

Bootstrapping Norwegian AMR for Parsing and Generation

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Abstract

This paper presents a methodology for adapting existing English AMR processing and generation models to Norwegian, a language previously unsupported. I fine-tuned the AMRLib library’s BART-based model for AMR parsing (Sentence-to-Graph) and their T5-based model for generation (Graph-to-Sentence) using custom Norwegian-AMR datasets. I detail the technical challenges encountered and the solutions implemented, including data formatting and dependency management. My results show that while the parsing model achieves a Smatch score of 59, and the generation model a non-zero BLEU score, the performance highlights the inherent difficulties of cross-lingual fine-tuning, particularly for the generation task. I analyze the results, compare them to other state-of-the-art approaches, and propose future work to improve performance.

1 Introduction

Abstract Meaning Representation (AMR) (Banarescu et al., 2013) is a semantic formalism that uses rooted, directed, labeled graphs representing semantics on a sentence level, meant to be easy for people to read, and easy for machines to traverse. In AMR graphs nodes represent concepts, and edges represent relations. The nodes may be either English words, e.g. *man*, Propbank frames (Kingsbury and Palmer, 2002), e.g. *come-01*, *meet-02* or other special keywords specific to AMR. An example of this can be seen in Figure 1. The goal of AMR parsing is to transform natural language text into AMR graphs, and the goal of AMR generation is to generate natural language from AMR graphs.

One central point to note about AMR is the lack of linguistic neutrality. AMR is first and foremost built with the intention of representing English language text, while also being largely centred around the also English-based Propbank frames. The creators of AMR have also made this intrinsic bias

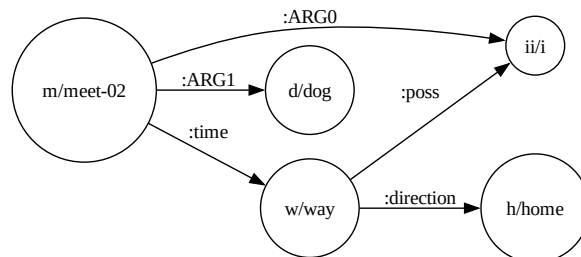


Figure 1: AMR graph for *I met a dog on my way home.*

clear. AMR is not meant to be an interlingua. Despite this, several efforts have been made to extend AMR parsing to other languages, evaluating its potential to work as an interlingua. The largest and most widely used manually annotated corpus for training AMR parsers and generators is the AMR 3.0 dataset (Knight et al., 2020), which has 59,255 hand-annotated AMR graphs along with their corresponding English sentences. More recently, other, larger datasets like MASSIVE-AMR (Regan et al., 2024) have been compiled, however these lack the quality control of AMR 3.0 and are often generated through machine translation.

AMR datasets with graphs in languages other than English have been made, most notably for Chinese (Li et al., 2021), but to a smaller extent, and most attempts at extending AMR parsing to other languages are still centred around using AMR graphs labelled in English, paired with sentences in the new language.

The primary hypothesis explored in this term paper is that the linguistic understanding of a model in one language can be effectively transferred to a new language with both a limited amount of target-language data, and a limited amount of supervision while generating this data. During this process I faced and overcame several technical challenges, including difficulties in managing versioning conflicts between core libraries, and correct preprocessing of data. The remainder of this paper is structured as follows: Section 2 discusses exist-

ing research on multilingual fine-tuning on AMR parsers. Section 3 details our fine-tuning approach and datasets used. Section 4 presents the results for both the parsing and generation models. Section 5 provides an analysis of the models’ performance, and Section 6 concludes with a summary and a discussion of additional work that could be done in the future.

2 Related Work

There have been two primary approaches in multilingual AMR parsing and generation to handle the scarcity of non-English data for training: building new models from the ground up for each target language, or adapting models already trained on English. The former approach usually relies on translating official AMR corpora issued by the LDC into the target language, or sometimes generating new corpora. The latter approach is often called cross-lingual transfer, and has become a popular approach due to how quickly parsers can be adapted to new languages with modern transformer models, even through unsupervised methods (Cai et al., 2021).

One notable example of the fine-tuning approach is Sheth et al. (2021). Their research explored the effectiveness of transfer learning for cross-lingual AMR parsing, by using an English AMR parser along with parallel corpora to train parsers for Spanish, German, Italian and Chinese. They showed that it is possible to effectively adapt parsers to new languages without the massive cost and effort associated with building gold-standard corpora from the ground up. This is also the approach in this term paper – given a sentence in Norwegian, reconstruct the AMR graph that was made for the English sentence the Norwegian one was translated from. Additionally, I trained a model for the inverse problem – constructing a Norwegian sentence corresponding to the English sentence an AMR graph was generated from.

Cross-lingual AMR parsing was also investigated by Vanroy and Van de Cruys (2024), where they parsed with BART-large models for English, Dutch and Spanish. They found that monolingual models tend to outperform multilingual ones, suggesting that if achieving state-of-the-art performance is the goal, it may be worthwhile to go through the effort of building gold-standard corpora from the ground up.

A middle ground was explored by Barta et al.

(2025), where they machine translated AMR 3.0 into Hungarian, quickly creating a silver-standard corpus, and then used it to train a Hungarian AMR parser. Additionally, development of new datasets, such as MASSIVE-AMR (Regan et al., 2024) provides new resources for languages previously unsupported, including Norwegian Bokmål. This however has a few problems, such as inconsistently translated entities, and a lack of standardised punctuation.

When it comes to AMR generation, it has been shown that it is possible to generate from English-centric AMRs into multiple languages by using multilingual pre-trained models, which further demonstrates the possibilities of transfer learning in this area. (Fan and Gardent, 2020)

3 Methodology

This section outlines the methodology used for building, fine-tuning, and evaluating my Norwegian AMR parsers and generator. The process involved model selection, data collection and formatting, setting up the AMRLib library along with its necessary dependencies, and lastly model training and testing.

3.1 Model Selection

For this project, I chose the AMRLib library as a framework to work through (Jascob, 2024). This decision was based on its reputation as a user-friendly and accessible tool, making it suitable for getting started with AMR parsing and generation. While getting it set up initially presented some challenges with conflicting dependency versions, these were resolved by using Python 3.9, first with a Docker container, then later by configuring a compatible environment with manually adjusted dependency versions. A key advantage of AMRLib is that it has several pre-trained parsing and generation models available for both direct use and further fine-tuning.

The specific model chosen for the parsing task was the `model_parse_bart_base-v0_1_0` model as a **Baseline Parser**. I then fine-tuned two versions of this model: the **Mixed-Norwegian Parser**, which was trained on a dataset of 400 Bokmål and the same 400 Nynorsk sentence-AMR pairs, and the **Bokmål Parser**, which was trained on an expanded dataset of 1,000 Bokmål pairs. Similarly, for the generation task, I used the pre-trained `model_generate_t5wtense-v0_1_0` model as my

Baseline Generator, and created a fine-tuned version on the same 1,000 Bokmål pairs, which I will call the **Bokmål Generator**. These models were chosen because of their high Smatch (82.3) and BLEU (44) scores in English AMR parsing and generation tasks, as well as for their speed. Other potential frameworks that were considered for this task included IBM’s Transition-AMR-Parser and Spring.¹

3.2 Data Collection and Formatting

Data for training and evaluation was sourced from the Tatoeba² parallel corpus, collected by manythings.org (Charles Kelly, 2025). This resource was particularly suitable because the sentences reflect a style of everyday conversational speech, in contrast to other parallel corpora that often contain long, complex sentences from legal or very formal texts. I decided to limit the corpus to including only sentences containing at least five words. The selected sentence pairs were between English and Bokmål, and then machine translated to include Nynorsk as well. Before a specific AMR parser library had been decided on, all initially selected sentences were tagged with numerical alignments between English and both variants of Norwegian, as per the approach in Sheth et al. (2021). These were however ultimately not used during fine-tuning, as they were not supported by any of the models included in the AMRLib library.

3.2.1 Mixed-Norwegian Parser

For fine-tuning the first parsing model I prepared data in both Bokmål and Nynorsk. The idea behind this was to test whether the resulting model would be able to parse both varieties of the language equally well given the same training data in both varieties. Initially 500 sentence pairs were gathered between Norwegian Bokmål and English to create a parallel corpus. To include Nynorsk, the Bokmål sentences were machine translated to Nynorsk using Apertium (Forcada et al., 2011). This was done since although Tatoeba has enough data between English and both Bokmål and Nynorsk individually, there was not a sufficiently large set of English sentences with translations in both Bokmål and Nynorsk. The translated Nynorsk sentences, along with the ones in Bokmål and English, were

then manually reviewed and corrected to fix any obvious translation errors.³ Due to time constraints, machine translation was used, although manual translation would have been an option.

The English component of each sentence pair was then processed using the Baseline Parser to generate AMR graphs. These generated AMRs were also manually inspected for quality, but no changes were ultimately made, despite occasional errors. One of erroneously parsed graphs is displayed in Figure 3.

The final dataset was then organised into plain text files split into train (800 pairs), test (100 pairs) and dev (100 pairs) sets in the PENMAN format, with each entry containing an AMR graph, tagged with its corresponding sentence and a unique ID. This format is illustrated in Figure 2.

```
# ::id tatoeba_nor.7
# ::snt Kvifor vil du snakka med meg ?
(c / cause-01
  :ARG0 (a / amr-unknown)
  :ARG1 (w / want-01
    :ARG0 (y / you)
    :ARG1 (t / talk-01
      :ARG0 y
      :ARG2 (ii / i))))
```

Figure 2: An AMR graph in PENMAN format with id and sent tags.

3.2.2 Bokmål Parser

After successfully creating the initial Mixed-Norwegian Parser, another Parser fine-tuned only on Bokmål was created, in order to try to achieve the best possible performance on Norwegian given the available resources allotted for this project.

Here the same 500 Bokmål sentence-AMR pairs from the previous model were used, expanded with another 700 pairs, split into sets of 1000 pairs for training, 100 for development, and 100 for testing.

3.2.3 Bokmål Generator

Subsequently the Baseline Generator was fine-tuned on the same Bokmål corpus used to fine-tune the Bokmål Parser. Although not the primary focus of this work, already having the necessary data to fine-tune such a model made it a fairly straight-forward

¹Alternative AMR parsing frameworks considered include <https://github.com/IBM/transition-amr-parser> and <https://github.com/SapienzaNLP/spring>.

²<https://tatoeba.org>

³For example, the Bokmål “Hvilket språk snakker de i USA?” (What language do they speak in the United States?) yielded “Noko som språk snakkar dei i USA?” in Nynorsk, and was corrected to “Kva språk snakkar dei i USA?”

expansion of the project, which allowed for comparison of the possible achievable performance for AMR Generation with that of Parsing.

3.3 Model Training

All model fine-tuning was done locally on a machine with no GPU, a process which took approximately one and a half hours per model, with a total of three models trained. As stated earlier, two distinct models were trained for the parsing task. The Mixed-Norwegian Parser was fine-tuned on the prepared data in PENMAN format from a txt file. Before fine-tuning, the hyperparameter fp16 in the model configuration had to be set to false as I had no GPU access. No further changes were made to the hyperparameters. The learning rate was at 5e-5 and an evaluation batch size of 32 was used, both values left at default.

Following this, the Bokmål Parser was trained using the expanded dataset of 1,000 sentence-AMR pairs. While training the Bokmål Parser a key modification was made to the hyperparameters: the number of warm-up steps was reduced from its default value of 3467 to 100. This default value was intended for a much larger training corpus, and with my limited corpus this meant the initial Mixed-Norwegian Parser never left the warm-up phase. I believe the change in this parameter, which was overlooked during training of the initial model, significantly influenced the final performance.

The generation model was trained following a similar procedure. This involved using the generated AMR graphs as input to produce corresponding Norwegian sentences. Slight configuration changes also had to be made to the generation model’s settings to successfully train on limited hardware. The checkpoint limit was changed from 1 to 2, the number of training epochs from 8 to 16, gradient accumulation steps from 2 to 1, and warm-up steps from 0 to 100.

4 Fine-tuning Results

In this section the results of the bootstrapped parsing and generation models are presented. I evaluate the performance of the models using standard metrics, including Smatch for AMR parsing and BLEU for AMR-to-text generation. I then go through the results to understand the models’ learned capabilities in Bokmål and Nynorsk and to identify the impact fine-tuning had on the baseline models.

4.1 Parsing Performance

In order to evaluate the effectiveness of the fine-tuning approach, the Baseline Parser, along with the Mixed-Norwegian and Bokmål Parsers were tested on a set of test sentences from English, Bokmål and Nynorsk. The results, measured by Smatch scores, are summarised in Table 1.

Parsing Model	ENG	NOB	NNO
Baseline	82.3	23.5	22.7
Mixed-Norwegian	-	38.1	37.7
Bokmål	-	58.9	53.3

Table 1: Smatch scores for the parsing models on different language test sets.

The data in the table shows the performance of the models across all three languages. The fine-tuning on Norwegian data significantly improves the model’s ability to parse these new languages compared to the zero-shot baseline trained only on English, even in the mixed model that was only fine-tuned on 400 unique AMRs.

The Smatch score on English for the two fine-tuned models could not be estimated, as I did not have access to a sufficiently large English-based test set. However, when parsing English, the Mixed-Norwegian Parser changed around 25% of resulting AMRs in some way, and the Bokmål Parser changed around 30% in some way. The changes were all minor, yet affected English parsing performance negatively.

4.2 Generation Performance

The Graph-to-Sentence Baseline Generator was fine-tuned on my set of 1000 Bokmål sentence-AMR pairs, and evaluated using BLEU scores. While fine-tuning the generation model it remained at a BLEU score of zero for most of training, though eventually achieved a non-zero score, and ended up with a BLEU score of 5.89.

5 Analysis and Discussion

This section goes through the results of the fine-tuning of all models, and discusses the results thereof.

5.1 Parsing Models

The following section provides an analysis of the parsing models’ performance, exploring the quantitative results and illustrative qualitative examples.

5.1.1 Quantitative Results

The Smatch scores from Table 1 clearly show that models fine-tuned on limited data are capable of considerable cross-lingual transfer, and improve significantly compared to the zero-shot performance of the Baseline Parser unfamiliar with Norwegian. We can also see that increasing the training data from 400 to 1000 unique AMRs boosts performance even more.

One interesting observation is that all parsers perform better on Bokmål than on Nynorsk; when trained on a balanced dataset of both like the Mixed-Norwegian Parser, or on neither of them like the Baseline Parser. This could potentially stem from that even though the BART model the parser is based on is supposed to be trained only on English, data from other languages may have leaked into its training. Of the two Norwegian varieties, Bokmål is by far the one with the largest online presence, and therefore more likely to leak into the training data of the BART model.

Another observation is that the Bokmål Parser was quite successful in generalising its Bokmål training data to parsing Nynorsk. Even when tested on Nynorsk it achieved a Smatch score roughly comparable to the one achieved when tested on Bokmål.

5.1.2 Qualitative Examples

One important thing to note is that none of my bootstrapped models were fine-tuned on gold-standard sentence-AMR pairs. They were all generated by the Baseline Generator from sentences taken from Tatoeba. That means they are limited not just by a lack of data for fine-tuning, but also the non-gold AMRs they are trained on, which to some extent will contain errors. Figure 3 contains one such AMR, where *seven* and *tomorrow* have been misinterpreted as a day and a month, respectively.

```
Please wake me up at seven tomorrow morning .
(w / wake-up-02
  :mode imperative
  :polite +
  :ARG0 (y / you)
  :ARG1 (ii / i)
  :time (d / date-entity
    :day 7
    :month (t / tomorrow
      :dayperiod (m / morning))))
```

Figure 3: An incorrect AMR graph parsed by the Baseline Parser.

```
Vanligvis kjøper jeg ikke noe på internett .
(p / possible-01
  :polarity -
  :ARG1 (g / go-02
    :ARG0 (ii / i)
    :ARG4 (h / house
      :poss ii))))
```

```
Vanlegvis kjøper eg ikkje noko på internett .
(p / possible-01
  :ARG1 (d / do-02
    :ARG0 (ii / i)
    :ARG1 (t / that)
    :time (d2 / day
      :mod (n / next)))
  :ARG1-of (c / cause-01
    :ARG0 (a / amr-unknown)))
```

```
Usually I do n't buy anything on the internet .
(b / buy-01
  :polarity -
  :ARG0 (ii / i)
  :ARG1 (a / anything)
  :medium (ii2 / internet)
  :mod (u / usual))
```

Figure 4: AMRs parsed by the Mixed-Norwegian Parser from a sentence in Bokmål and Nynorsk respectively, with the correct AMR below.

In figure 4 we can see that even when equivalent Bokmål and Nynorsk sentences contain only minor differences, the resulting AMRs can end up quite different even when generated by the Mixed-Norwegian Parser. This is not entirely unexpected when the model has so little fine-tuning data to base its predictions on, but still surprising considering that the fine-tuning data consisted of equivalent Bokmål and Nynorsk sentences, represented by the same AMR graphs.

5.2 Generation Model

The following section provides an analysis of the generation models' performance, exploring the quantitative results and illustrative qualitative examples.

5.2.1 Quantitative Results

The fine-tuned generation model's BLEU score of 5.89 is low compared to the non-finetuned model, which was at 44 before Bokmål finetuning data was introduced. The generation model was much less capable of adapting to the fine-tuning data than the parsing model. This is however to be expected, as the output for the bootstrapped parsers remained AMRs, a format they were familiar with, while the output for the generation model changed from English to Bokmål, an entirely new language, which it was not trained on until the introduction of the very

limited dataset of 1000 fine-tuning sentence-AMR pairs.

The generator was however able to unlearn generating English, and change to generating something that is recognizable as an attempt to imitate Norwegian.

5.2.2 Qualitative Examples

```
ENG: The capital city of the Netherlands is Amsterdam .
NOB: Hovedstaden i Nederland er Amsterdam .
GEN: Amsterdam er en holländische kapitale .
(h / have-org-role-91
  :ARG0 (c / city
    :name (n / name
      :op1 "Amsterdam"))
  :ARG1 (c2 / country
    :name (n2 / name
      :op1 "Netherlands"))
  :ARG2 (c3 / capital))
```

Figure 5: A sentence generated from an AMR by the fine-tuned generation model.

```
ENG: You need a license to drive a car .
NOB: Du trenger førerkort for å kjøre bil .
GEN: Du m lre t kunne kjrer .
(n / need-01
  :ARG0 (y / you)
  :ARG1 (l / license-01
    :ARG1 (d / drive-01
      :ARG0 y
      :ARG1 (c / car))))
```

Figure 6: A sentence generated from an AMR by the fine-tuned generation model.

In Figure 5 and 6 we see examples of sentences generated from AMRs by the Bokmål Generator, on the line marked with *GEN*:. Most generated sentences are comparable to these, where it is possible to see a certain similarity between the original sentence and the one generated.

A notable finding in the Bokmål Generator’s output, seen in Figure 5, was the occasional appearance of German words, something which was not observed in output from the Baseline Generator. This could potentially be explained by the nature of the underlying T5-based model, which is a pre-trained multilingual transformer. My fine-tuning on Norwegian seems to have activated this broader linguistic knowledge. German is the only language that was observed to start appearing after fine-tuning, no interference from other languages seems to be present, apart from a handful of English words. We also see that *capital* is generated

as *kapitale*, which could indicate that the model has learned that Norwegian generally does not use the letter *c* (except for in recent loans not yet integrated into the spelling), replacing it with *k* in words that otherwise correspond closely to their English equivalent.

Additionally it can be noted that the Norwegian letters *æ*, *ø*, and *å* are not generated by the model, and instead skipped entirely. Therefore, in Figure 6 the words *m*, *lre*, and *kjrer* are probably attempts at generating *må*, *lære*, and *kjører*. Even though the words *må* (*need to*), *lære* (*learn*), and *kunne* (*be able to*) do not occur in the original sentence the AMR is generated from, they do not seem entirely out of place in the given context.

5.2.3 Discussion of Results

Analyzing the output of the fine-tuned models demonstrates that cross-lingual transfer is viable for both AMR parsing and generation in Norwegian. The parsing task was highly successful, boosting the Smatch score significantly above that of the baseline, while the influence of pre-training data biases can also be seen from the persistent performance gap between Bokmål and Nynorsk. The Smatch scores should however only be accepted with caution; both training and testing data were taken from the same source, with all sentences being roughly the same (short) length. This could have artificially boosted the Smatch score. If the models had been tested on more diverse data the scores observed may have ended up lower.

The generation task was more challenging, yielding a low BLEU score, though it successfully produced Norwegian-like output. The appearance of German words in the output gives us revealing insights into the model’s linguistic latent space and the limitations of fine-tuning with a small dataset. Such observations also help us see the challenges of adapting multilingual models to previously unsupported languages.

Analysing the output of the fine-tuned models reveals two key insights. First, the parsing task proved to be highly suitable for cross-lingual transfer, with both fine-tuned models demonstrating a significant leap in performance over the zero-shot baseline. The Smatch scores clearly show that even with limited non-gold training data, the underlying model successfully learned to adapt to parsing Norwegian syntax and mapping the language to English AMR structures. The observed performance difference between Bokmål and Nynorsk highlights the

persistent influence of pre-training data, suggesting that the model’s potential greater exposure to the more prevalent Bokmål biased its performance despite balanced fine-tuning data. It should however still be noted that the Bokmål parser proved fairly capable of transferring its understanding of Bokmål over to parsing Nynorsk.

Second, the generation task, while being more challenging due to the complete change in output language, still gave results. The low BLEU score is a product of the model’s struggle to adapt from English to Norwegian without a larger corpus, yet it still successfully learned to produce output that is recognizably Norwegian, and that somewhat relates to the context described by the AMR. The appearance of German words in the output shines light on the model’s multilingual knowledge of related languages being able to lead to subtle but revealing errors. These findings collectively suggest that while fine-tuning on a small dataset can effectively bootstrap parsing and generation capabilities in a new language, the inherent biases and complexities of multilingual models remain a significant factor to consider.

6 Conclusion and Future Work

In this term paper, I successfully fine-tuned a pre-trained multilingual transformer for both AMR parsing and text generation in Norwegian. The key findings demonstrate that even with a limited, non-gold-standard dataset, the models were capable of significant cross-lingual adaptation. The parsing models achieved a substantial performance increase over the zero-shot baseline, in addition to an observed performance bias towards Bokmål underscoring the impact of the model’s pre-training data.

While the generation task proved more challenging, and definitely could have benefited from more fine-tuning data, as evidenced by the low BLEU score and occasional cross-lingual interference from German, it successfully adapted the model’s output from English to a recognizable form of Norwegian. It must also be noted that the purpose of such a model is not to adapt it to a completely alien output format, which is what was done in this project.

Like other research has proven earlier (Sheth et al., 2021) these results confirm the potential of this methodology for bootstrapping semantic resources for new languages, even in a limited ap-

proach such as this one, but also highlight the complexities that come from the underlying nature of such models.

Building on these findings, future work could explore several directions. Other related work has used significantly larger datasets for fine-tuning, several tens of thousands of sentence-AMR pairs compared to the 1,000 pairs used in my largest model. This was not possible in this work due to time and resource constraints, but certainly would have yielded better results. A deeper error analysis, similar to that in (Sheth et al., 2021) could also reveal more of the inherent flaws in this approach, which would give more insight into how these could be combatted in future approaches.

A secondary dataset containing word-level alignments between English and the two variants of Norwegian was collected, but not used in this work. Finding a way of incorporating these alignments into a future approach could potentially significantly increase the accuracy of both the parsing and generation tasks.

Additionally, it would be beneficial to investigate the impact of specific hyperparameter changes in more depth, such as a different warm-up schedule or learning rate, to see how this affects model convergence and final performance. Looking into other model architectures, or implementing a more robust evaluation metric for earlier stages of training could also give a more comprehensive understanding of the various challenges and opportunities in the domain of bootstrapping AMR for parsing and generation.

Acknowledgements

I would like to thank Johannes Dellert for giving me the idea for this project, and for his guidance throughout it.

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